Verification and validation of axisymmetric solvers for hole opening in a viscous sheet

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Abstract

We document four solvers for the axisymmetric Taylor–Culick problem — the opening of a circular hole in a viscous sheet — and the evidence that each solves the equations it claims to. Two are depth-averaged: a Lagrangian per-radian mass coordinate carrying the radial metric as an unknown, and an arbitrary Lagrangian–Eulerian (ALE) solver in the physical radius. Two resolve the free surface directly: a laboratory-frame calculation on a graded annulus, and a co-moving window that translates outward with the rim. The depth-averaged equations were derived independently and agree term for term with Savva & Bush's (3.15)–(3.16); the direct formulations were audited line by line against two independent implementations of the axisymmetric operators. In this geometry a radial translation is not a Galilean boost, so the co-moving trick that serves the planar sheet has no analogue: the laboratory frame is limited by the finite sheet it must mesh, and the window formulation, not a change of variables, is what represents a semi-infinite sheet. Verification is by an internal-check ladder, whose gate is the planar limit: at large initial radius R_0 both depth-averaged solvers reproduce the validated planar solver with a deviation falling as $0.37/R_0$, under the $2/R_0$ admission bound at every rung, and the direct solver reproduces the validated planar retraction speed to 0.13% at $R_0 = 100$. Against the 2024 volume-of-fluid campaign the direct solver agrees to 1.6% in radius and 1.8% in speed, systematically and at face value. The depth-averaged model at $R_0 = 1$ does not reproduce the direct solution: the two available tip closures bracket it from above and below, 0.9966 and 0.8976 against 0.9613 at $t^* = 100$, which locates the discrepancy in the edge condition rather than in the bulk. Four numerical traps found in the course of this work are recorded with the evidence that identified each, because three of them return a plausible wrong answer or a misleading failure rather than an error.

1 Scope

This note covers the Newtonian axisymmetric strand: four solvers, the checks each passes, and the small number of end-to-end comparisons made against external data. It is a companion to the planar validation note, and it inherits that note's conventions, its planar solvers as references, and its warning about Ohnesorge conventions.

Two limits on what follows should be stated at the outset rather than discovered in §5. First, the comparisons in §5 are one end-to-end run per question. This strand exists to establish that the axisymmetric machinery is correct and that its answers are not obviously wrong, not to benchmark it exhaustively; a sweep in Oh and R_0 is deliberately out of scope. Second, the internal checks of §4 test discretisations against the equations and against each other. They are not evidence that the depth-averaged model describes a real opening hole, and §5 shows a case in which it plainly does not.

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1.1 Nomenclature and units

The non-dimensionalisation is unchanged from the planar work: $\rho = \sigma = h_0 = 1$ with h_0 the *full* sheet thickness, so the half-thickness is 1/2 and the Taylor–Culick speed [1, 2] is $V_{TC} = \sqrt{2\sigma/\rho h_0} = \sqrt{2}$. Throughout, $\mathrm{Oh} = \mu/\sqrt{2h_0\rho\gamma}$ is the Savva & Bush Ohnesorge number [3], the coded viscosity is $\mu = \sqrt{2}\,\mathrm{Oh}$, the viscous clock is $\tau_{\mathrm{vis}} = \mu/2$ and $t^* = t/\tau_{\mathrm{vis}}$. The reported speed is $u^* = \dot{R}/V_{TC}$, which tends to unity.

One parameter is new: the initial hole radius R_0, in units of h_0. The combination that governs the early behaviour is $R_0^* = R_0/(\mathrm{Oh}\,h_0)$, the Savva & Bush similarity variable [3]; $R_0^* \ll 1$ is the small-hole regime in which the hole grows exponentially and $R_0^* \gg 1$ is the regime that starts like the planar sheet. It is worth computing before choosing a test, because neither closed form is a clean target in between.

All four are built on the same symbolic finite-element framework [4], which derives residuals and Jacobians from the weak forms written below. Their entry points; all four modules live in `simulationCases/`:

name	module	what it is
TC1D-AXI-MASS	`trouton1d_axi.py`	depth-averaged, per-radian Lagrangian mass coordinate, fixed domain, radial metric carried as an unknown
TC1D-AXI-ALE	`trouton1d_ale_axi.py`	depth-averaged, physical radius on a moving mesh
AXI-LAB	`axisymmetric_lab.py`	free-surface Navier–Stokes, laboratory frame, graded annulus
AXI-COMOVING	`axisymmetric_comoving.py`	free-surface Navier–Stokes, window of fixed length translating with the rim

2 The depth-averaged axisymmetric model

The guiding choice is a *metric* formulation. The grids stay one-dimensional exactly as in the planar codes; axisymmetry enters through the r-weighted measure and the hoop terms of the governing equations, and not through a second spatial dimension. Everything below was re-derived from first principles and then checked term by term against Savva & Bush's equations [3]; the planar limit reproduces the residuals of the planar modules exactly, which is what the check of §4.3 measures.

2.1 Membrane form

Let r be the radial coordinate, $R(t)$ the hole edge, and let the film occupy $r > R$. With $u(r,t)$ the depth-averaged radial velocity and $h(r,t)$ the full thickness, the plane-stress ($\sigma_{zz} = 0$) viscous membrane stresses, with the extensional response of Trouton [5], are

$$\sigma_{rr} = 2\mu\left(2u_r + \frac{u}{r}\right), \qquad \sigma_{\theta\theta} = 2\mu\left(u_r + 2\frac{u}{r}\right), \qquad N_{rr} = h\sigma_{rr}, \quad N_{\theta\theta} = h\sigma_{\theta\theta}, \tag{1}$$

and the depth-averaged mass and momentum balances are

$$\frac{\partial h}{\partial t} + \frac{1}{r}\frac{\partial}{\partial r}\left(ruh\right) = 0, \tag{2}$$

$$h\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial r}\right) = \frac{\partial N_{rr}}{\partial r} + \frac{N_{rr} - N_{\theta\theta}}{r}. \tag{3}$$

Equation (2) is Savva & Bush's (3.15). Expanding the right-hand side of (3) gives

$$4\nu h\left(u_{rr} + \frac{u_r}{r} - \frac{u}{r^2}\right) + 2\nu h_r\left(2u_r + \frac{u}{r}\right), \tag{4}$$

which is their (3.16) multiplied through by h; the identity was verified by expanding their form $(4\nu/h)\left[((h/r)(ur)_r)_r - uh_r/(2r)\right]$ and matching term for term. Two terms are new relative to the planar model: the metric weight in the mass flux and in the stress divergence, and the hoop stress $N_{\theta\theta}/r$. Both vanish as $h_0/R \to 0$, so the planar operator is the exact $R_0 \to \infty$ limit — which is a statement the code can be made to prove, and §4.3 does.

The distributed capillary term of Savva & Bush's (3.16) stays *off* by default, exactly as in the planar solvers: the tip closure below already carries the capillary driving, so a bulk curvature term would double-count it. For a cell-constant thickness the elementwise gradient of h is identically zero, so switching bulk capillarity on is not a flag but a change of variable — it would have to arrive together with $f = h^2$, whose curvature is the only form bounded at the tip [3]. The solver rejects the combination rather than integrating a silently vanishing term.

2.2 Mass-coordinate form, and the metric field $w = r^2/2$

With the per-radian mass coordinate $\mathrm{d}m = rh\,\mathrm{d}r$ the tip sits at $m = 0$ and the domain $[0, M]$ is fixed, which is the property that makes this the accurate late-time formulation. The price is that $\partial_r = rh\,\partial_m$ carries the radius, so the radial map appears in every residual and must be carried as an unknown.

The choice made here is to carry $w = r^2/2$ on continuous piecewise-linear elements rather than r itself, constrained by

$$\frac{\partial w}{\partial m} = \frac{1}{h}, \qquad r = \sqrt{2w}, \tag{5}$$

together with one global scalar $R(t)$ satisfying $\dot{R} = u|_{m=0}$ and $w|_{m=0} = R^2/2$. Three properties recommend (5) over the obvious alternatives. It is *linear* in the carried unknown, where $r_m = 1/(rh)$ is not. With w piecewise linear and h cell-constant both sides are cell-constant, so (5) is a per-cell statement and the discrete radial map is exactly consistent with the discrete mass distribution, cell by cell, to round-off — a property that is measured rather than assumed (§4.1). And $r^2 = 2w$ exactly, so every even power of the radius in the residuals is a polynomial in the unknown and only the odd powers cost a square root. Evolving $r_t = u$ pointwise instead is consistent only in the continuum: it lets the discrete radial map drift away from the discrete mass it is meant to describe.

The momentum weak form is written so that the only reciprocal anywhere in the residual is a single guarded $1/r^2$. With test function v and a Cartesian measure in m,

$$\int u_t v\,\mathrm{d}m + \int \left[4\mu r^2 h^2 u_m + 2\mu h u\right] v_m\,\mathrm{d}m + \int \left[2\mu h u_m + \frac{4\mu u}{r^2}\right] v\,\mathrm{d}m - 2R\,v(0) = 0. \tag{6}$$

Coding the hoop term literally as $\sigma_{\theta\theta}/r$ instead introduces a spurious $1/r$ that cancels analytically but not numerically. That is the restructuring a future refactor is most likely to reintroduce, and it is why the dissipation check of §4.1 exists: setting $v = u$ collapses the viscous terms to $\int 4\mu\left[u_r^2 + u_r(u/r) + (u/r)^2\right]\mathrm{d}m$, a positive-definite form with eigenvalues $1/2$ and $3/2$, so a sign error in the hoop term drives it negative immediately.

Discretisation follows the planar production choice: h cell-constant, u and w continuous piecewise linear, no midside velocity degrees of freedom anywhere, BDF2 in time. This is the finite-element realisation of the staggered arrangement Savva & Bush adopt, following Eggers & Dupont [6], for the same reason: they report that a non-staggered grid introduced spurious oscillations in the curvature at long times [3]. The co-located quadratic arrangement is retained in the planar module for reproducing older runs and is deliberately unavailable here, because it is non-convergent.

2.3 ALE form

TC1D-AXI-ALE integrates (2)–(3) in the physical radius on $[R(t), R_{\mathrm{out}}]$, with the mesh moving under a Laplace-smoothed mesh equation. In one dimension that smoothing is the affine stretch

$x(X) = R(t) + (R_0 + \ell_0 - R(t))\,(X - R_0)/\ell_0$, which is precisely the mapping Savva & Bush use [3], applied to the offset domain.

Here the measure does the work. Selecting the axisymmetric coordinate system attaches the factor $2\pi r$ to *every* integral the framework assembles on the domain, including the zero-dimensional point contribution a boundary condition makes at a mesh endpoint. Integrating (3) by parts under that measure leaves the full N_{rr}, including its u/r part, in the flux slot, and the hoop term outside it as

$$\underbrace{\int N_{rr}\, v_r\, (2\pi r)\, \mathrm{d}r}_{\text{flux slot}} \qquad \text{and} \qquad \underbrace{\int \frac{N_{\theta\theta}}{r}\, v\, (2\pi r)\, \mathrm{d}r = \int N_{\theta\theta}\, v\, (2\pi)\, \mathrm{d}r}_{\text{hoop term}}. \tag{7}$$

The $1/r$ written explicitly in the hoop term is therefore not a numerical convenience; it is exactly what cancels the extra radius the measure would otherwise leave on that term. This form was verified by expansion rather than assumed, because it is the one place where the automatic measure and the hand-written residual have to agree.

Both fields are continuous piecewise linear here. The mass equation carries an advective term in this frame, so a cell-constant thickness would need an upwind flux; co-location is safe because the parasitic mode that rules it out in the planar mass-coordinate solver lives on midside nodes, which a linear line mesh does not have.

One property of the affine map is a known limitation rather than a defect. It compresses resolution uniformly across the domain as R grows, so it loses near-tip resolution by orders of magnitude if R itself grows by orders of magnitude — which is the small-hole exponential regime. An area- or arclength-preserving stretch is the recorded upgrade path.

2.4 Tip closure, and an optional geometric correction

At leading order the closure is unchanged from the planar case: the massless-cap force balance $N_{rr}(R) = -2\gamma_{\text{total}} = -2$. Savva & Bush's axisymmetric jump condition (3.20) is the same $[u_r] = -1/4$ as their planar (2.22), so the constant is the planar one. What the geometry adds is supplied by the measure: under an r-weighted measure the boundary term automatically carries the factor $r_{\text{tip}} = R$, giving the correct per-radian pull $2\gamma R$. In the mass-coordinate solver this makes the boundary value nonlinear in a global unknown, which the symbolic Jacobian handles without special-casing.

The first geometric correction is available behind a flag and is off by default. The toroidal rim surface opposes opening with a force $\gamma\pi h_{\text{tip}}/(2R)$ per unit edge length, so

$$N_{rr}(R) = -2 + \frac{\pi h_{\text{tip}}}{2R}. \tag{8}$$

It was derived two ways — as the surface-energy release rate of the cap, and by integrating the azimuthal Laplace pressure γ/R over the cap arc $\pi h_{\text{tip}}/2$ — and it is the $O(h/R)$ effect that the early-time analysis of Savva & Bush neglects, their scaling requiring $h_0/R_0 \ll 1$. At $R_0 = h_0 = 1$ that correction is $O(1)$ at $t = 0$. The flag therefore exists to *measure* the term's importance against a direct calculation, not to assert that it belongs in the leading-order model; §5.3 is that measurement.

2.5 Why $R_0 = 0$ is excluded

A vanishing initial hole is excluded on four independent grounds, and it is worth recording all four because any one of them alone reads as a numerical inconvenience. The early similarity solution $R(t) = R_0 \exp(t/2\tau_{\text{vis}})$ is linear in R_0, so a zero hole never opens. The mass map degenerates, $\mathrm{d}r/\mathrm{d}m \to \infty$. The edge force and the rim mass both vanish, leaving the tip acceleration a $0/0$. And the azimuthal curvature $1/R$ diverges: below the Taylor & Michael threshold at $R \sim h_0$ a

hole closes rather than opens [7]. Production depth-averaged runs use $R_0 \geq 5$, which is Savva & Bush's own smallest; $R_0 \sim 1$ sits at about twice the closing threshold and is reserved for the comparison of §5.3, where it is the direct solver's initial condition that forces the choice.

3 Two direct formulations, and the frame question

3.1 The laboratory frame on a graded annulus

AXI-LAB solves the incompressible Navier–Stokes equations in the liquid only, with a true free surface, on an annular domain $r \in [R(t), R_{\text{out}}]$ with the midplane $z = 0$ a symmetry line. The hole behind the rim is not meshed, so the liquid domain *shrinks* as the hole opens. The mesh is graded to the current tip and coarsens to the quiescent far film, which needs only about two elements across the half thickness.

The axisymmetric coordinate system is already the metric formulation: the mesh stays two-dimensional in (r, z), the measure gains $2\pi r$, the operators gain their hoop terms, and the free-surface condition in surface-divergence form carries *both* principal curvatures with no curvature code written anywhere. That last point is the one worth checking rather than trusting, and the static-droplet discriminator of §4.4 checks it.

The initial geometry is the annular translate of the planar one: tip at $(R_0, 0)$, a cap semicircle of radius $1/2$ centred at $(R_0 + 1/2, 0)$, and the film running out to R_{out}. At $R_0 = 1$ this matches the initial condition of the 2024 volume-of-fluid campaign [8] exactly, which is what makes the comparison of §5.2 like for like. The hole radius $R(t)$ is a pure observable — the minimum radial mesh coordinate on the free surface — and $\dot{R}$ follows by differentiation. Because no fluid crosses any boundary, exact global volume conservation is available free as a sharp correctness diagnostic, and it is monitored on every output.

3.2 A radial translation is not a Galilean boost

In the planar problem the retraction can be computed in a frame that follows the rim: freeze the tip, blow the far film in at $-U$, promote U to a global degree of freedom. That trick has no axisymmetric analogue. A radial translation of an axisymmetric geometry is not a Galilean boost, so there is no change of variables in which the far film is uniformly in motion and the governing equations are unaltered. Two consequences follow, and both are structural rather than matters of taste.

First, AXI-LAB carries no global frame-velocity unknown, no bulk force for frame acceleration, and no tip-pinning constraint — because none of that machinery exists, none of the framework's recorded traps can fire in it. Second, the fixed outer wall at R_{out} must contain the whole trajectory, so the reachable end time is bounded by the mesh budget rather than by the physics. That is the honest limitation of the laboratory frame, and it is the reason for the second direct solver.

3.3 The co-moving window

AXI-COMOVING keeps the *window length* L fixed and drags it outward with the rim, so the degree-of-freedom count is independent of the end time. This is an ALE window translation, not a change of variables: the coordinate system remains axisymmetric, the mesh coordinates carry the true radius, and the velocities are laboratory-frame throughout. It is the proper semi-infinite-sheet formulation of this problem, and it is not equivalent to any co-moving frame.

Three consequences are worth stating. There is no non-inertial correction, and porting the planar one would be a physics error, not a conservative choice. The far boundary is a genuine inflow of quiescent liquid. And volume is *not* conserved, so the free diagnostic of the laboratory case is replaced by an exact modified identity: the volume at time t must equal the initial volume plus the liquid the advancing far edge has swept in, $V(t) = V(0) + \frac{\pi}{2}\left[(X + L)^2 - (X_0 + L)^2\right]$

for a film of half-thickness 1/2, with X the window position. That identity is checked at every output.

The retraction speed here is a solved unknown rather than a post-processed difference. A second global degree of freedom U satisfies $\dot{X} - U = 0$ and is differentiated by the same BDF2 scheme as everything else, instead of by a first-order difference across a geometrically stretching output schedule. The driver logs both, and their difference is a free accuracy monitor on the output cadence.

4 Verification

Each solver carries a check suite that runs it through the same driver the campaign uses, so the checks exercise the production path rather than a bespoke harness. The suites are

solver	check suite
TC1D-AXI-MASS	`validation/trouton1d_axi_checks.py`
TC1D-AXI-ALE	`validation/trouton1d_ale_axi_checks.py`
AXI-LAB	`validation/axi_smoketests.py`

The ordering within a suite is chosen for cost, not for logic. The gate is the planar limit: it is the only check that compares against an independently validated solver rather than against the formulation's own internal structure, and until it passes nothing else in the suite is evidence of correctness. A metric constraint can be satisfied exactly by a wrong metric, and a positive-definite dissipation stays positive under a wrong prefactor.

4.1 The mass-coordinate solver

Metric consistency. The per-cell residual of (5) is $|w_m - 1/h| = 5.8 \times 10^{-14}$ at worst over all cells and all outputs. This is the structural payoff of the staggering, and if it ever drifts the discretisation of w is wrong.

Metric closure. The global identity $r(M)^2 = R^2 + 2\int_0^M \mathrm{d}m/h$ holds to 7.5×10^{-16} relative. Mass itself is exact by construction in this coordinate, since M does not move; the metric identity is what stands in its place.

Tip-thickness identity. The closure implies exactly $\mathrm{d}h_{\mathrm{tip}}/\mathrm{d}t = 1/(2\mu) - h_{\mathrm{tip}}u_{\mathrm{tip}}/(2r_{\mathrm{tip}})$. Integrating that ordinary differential equation alongside the run, using the solver's own u_{tip} and r_{tip}, the solver's first-cell thickness matches it to 6.7×10^{-4} relative. The planar limit of the same identity recovers $h_{\mathrm{tip}} = 1 + t^*/4$; its late-time axisymmetric limit is $h_{\mathrm{tip}} \to t^*/6$, azimuthal stretching thinning the rim by exactly 2/3 relative to planar. That last result was derived independently twice and is the headline non-planar prediction of the reduced model.

Early exponential growth. In the small-hole regime, at Oh = 100 and $R_0 = 5$ so that $R_0^* = 0.05 \ll 1$, the fitted logarithmic growth rate of $R(t)$ is within 1.04% of $1/(2\tau_{\mathrm{vis}})$, which is Savva & Bush's (3.30). The check reports the rate over four fit windows rather than one, so the sensitivity to that choice is visible rather than hidden inside a pass. The similarity solution is an early-time law that an impulsive start has to settle onto, so the first fraction of a viscous time is a start-up transient with a rate below the asymptote, and the window has to be chosen to exclude it.

Dissipation. The depth-averaged dissipation density is non-negative at every cell and every output, and finite everywhere. Alongside it the discrete energy budget $\mathrm{d(KE)}/\mathrm{d}t = 2Ru_{\mathrm{tip}} - D$ is reported, with a worst relative residual of 5.1×10^{-3}; that number is limited by the finite-difference time derivative taken on a geometric output cadence, not by the balance, and it is reported rather than gated for that reason.

4.2 The ALE solver

Mass conservation. The relative drift of $2\pi \int hr\,\mathrm{d}r$ is 5.8×10^{-9} at 2000 elements. The drift is spatial-error dominated: doubling the number of timesteps at fixed mesh does not move it, whereas the mesh ladder gives 3.6×10^{-8}, 9.1×10^{-9} and 5.8×10^{-9} at 800, 1600 and 2000 elements.

Mesh affinity. The mesh sits on the affine stretch of §2.3 to $\sim 10^{-14}$ of the domain length, so the mesh-motion solution is verified rather than assumed.

Early exponential growth. As above, within 3% of $1/\mu$ over $t^* \leq 2$ at $\mathrm{Oh} = 100$, $R_0 = 5$ ($R_0^* = 0.035$). The window is justified by where the regime ends: the exponential law breaks down once R approaches $\mathrm{Oh}\, h_0$, here at $t^* \approx 6$, and $R(t^* = 2) \approx 13.6$ is comfortably inside it.

Cross-solver. At $\mathrm{Oh} = 1$, $R_0 = 5$, the two depth-averaged solvers agree to $\max|\Delta u^*| = 7.3 \times 10^{-6}$. They share the equations and nothing else: a Lagrangian mass coordinate with a carried metric against a physical radius on a moving mesh, a cell-constant thickness against a continuous one, a nonlinear-in-R boundary value against a measure-supplied one. That agreement is the strongest internal evidence available for either of them.

4.3 The planar limit, which is the gate

Both depth-averaged solvers are compared against the validated planar solver at matched Oh, matched end time, matched timestep and matched output schedule, so the comparison is row by row with no interpolation. The meshes are matched by *tip cell in radius* and not by element count: the axisymmetric domain is a mass, so an equal element count would place a systematically different cell at the tip and contaminate the difference with a discretisation offset that does not scale with $1/R_0$.

R_0	50	100	200	400
TC1D-AXI-MASS, $\max\|\Delta u^*\|$	3.355×10^{-3}	2.717×10^{-3}	1.675×10^{-3}	9.246×10^{-4}
TC1D-AXI-MASS, $R_0 \max\|\Delta u^*\|$	0.168	0.272	0.335	0.370
TC1D-AXI-ALE, $\max\|\Delta u^*\|$	3.357×10^{-3}	2.718×10^{-3}	1.675×10^{-3}	—
TC1D-AXI-ALE, $R_0 \max\|\Delta u^*\|$	0.168	0.272	0.335	—
bound $2/R_0$	4.0×10^{-2}	2.0×10^{-2}	1.0×10^{-2}	5.0×10^{-3}

Every rung is inside the $2/R_0$ bound with an order of magnitude to spare, and the scaled deviation $R_0 \max|\Delta u^*|$ rises towards a constant 0.37 ± 0.03 rather than towards zero or towards infinity, which is the statement that the leading correction is $O(1/R_0)$. The observed order in $1/R_0$ improves down the ladder, 0.30, 0.70, 0.86, reaching 0.93 on a further $R_0 = 800$ probe. The approach is one-sided and slow for a reason that is worth recording: the sub-dominant term in the difference is governed by L/R_0 with L the domain length, not by the small parameter h_0/R_0, so linearity is asserted only on the asymptotic pair while the magnitude bound is asserted at every rung. Figure 1(b) shows the ladder.

The ladder deliberately stops at $R_0 = 400$–800. At $R_0 = 10^6$ the flux slot of (6) would carry $r^2 h^2 u_m$ with scale factors of order 10^{12}, and the measurement would be of conditioning rather than of the metric terms.

4.4 The direct solvers

Hoop-curvature discriminator. A static quarter-disc droplet of radius a is solved to stationarity and its volume-averaged pressure compared with Young–Laplace. The axisymmetric answer is $2\sigma/a$ and the planar answer σ/a, so the factor of two is a clean discriminator that the azimuthal curvature is genuinely active rather than the coordinate system merely being labelled axisymmetric. Measured over a resolution ladder the exponent comes out at $p = 1.9997$–1.9999 against 2.

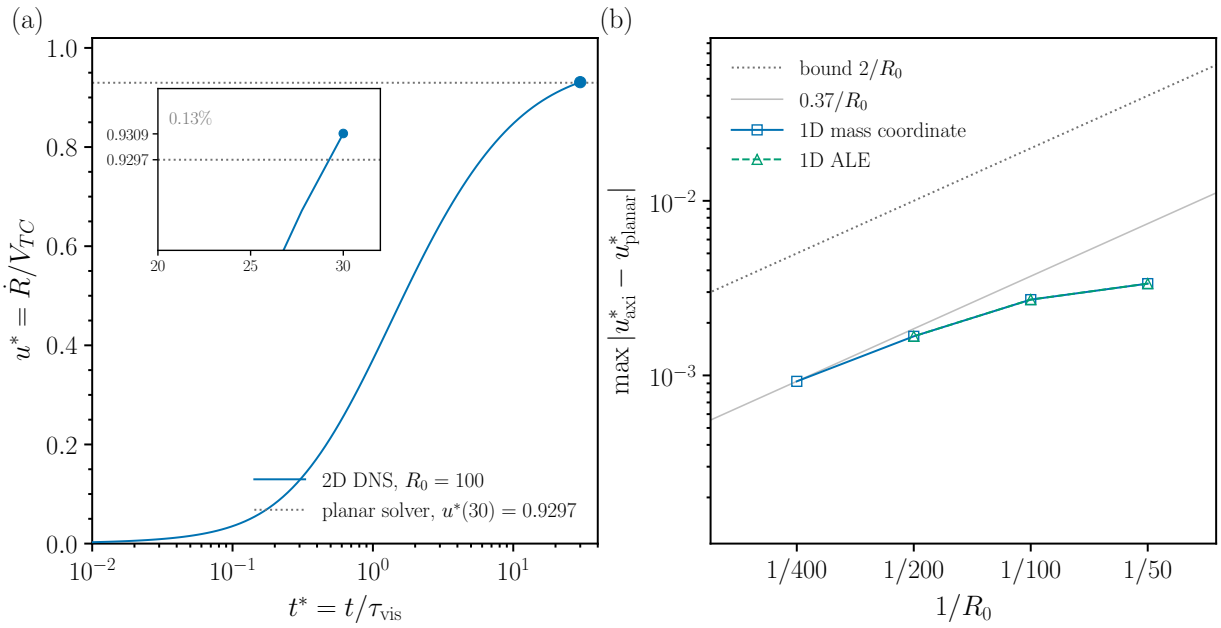


Figure 1: The planar-reduction gate. (a) The direct solver at $R_0 = 100$, Oh $= 0.5$, against the validated planar value $u^*(t^* = 30) = 0.9297$; the inset resolves the 0.13% gap that the main axes cannot show. (b) The depth-averaged planar-limit ladder at Oh $= 1$: squares TC1D-AXI-MASS, triangles TC1D-AXI-ALE, dotted the $2/R_0$ admission bound, grey the line $0.37/R_0$. The two depth-averaged solvers are indistinguishable at this scale — they agree to four significant figures at every rung — so the two symbols overlie each other; that agreement is the cross-solver check of §4.2 seen from a second direction.

This check has a trap of its own, recorded because it produced a converged and meaningless answer. Without mesh position degrees of freedom the free-surface condition resolves to its static branch, and the kinematic multiplier then enforces $\mathbf{u} \cdot \mathbf{n} = 0$ as a rigid-wall reaction, absorbing the entire surface-tension load itself and leaving the bulk pressure at solver noise. The residual converges to machine precision in one Newton step; the system is simply rank-deficient for the pressure. Adding the smoothed mesh routes the multiplier into the mesh-position equation instead, which changes which equation the multiplier feeds and not the physical answer.

Jacobian. The analytic Jacobian agrees with a finite-difference Jacobian over several steps of the real case at coarse resolution, with no aborts.

Volume. Relative drift is 2.1×10^{-9} over 200 steps with remeshing suppressed, and about 2.6×10^{-5} over a full remeshing run. The difference is the interpolation error of the remesh operations, not accumulation, and is discussed in §7.

Resolution ladders. At $R_0 = 1$ the resolution knob was itself checked before being used, because the total element count is insensitive to it on a large domain and that insensitivity had been read as the knob doing nothing. The minimum interface edge length at the tip tracks the requested resolution to within 0.5% of the ratio (2.00 for a factor-of-two change), and the near-tip element count scales as the analytic grading law predicts; what goes flat is the total count far from the tip, so counting elements over the whole annulus is simply not a valid diagnostic here. On that verified ladder, halving the tip edge length moves $R(t^* = 50)$ by 5×10^{-7} relative. Doubling the output density — which is what sets the timestep — moves $R(50)$ by 1.9×10^{-6} and $\dot{R}(50)$ by 4.3×10^{-4}.

Window against wall. The co-moving and laboratory solvers were run on the same physical problem over an early window at resolution 0.06, and their hole radii agree to $|R_{\mathrm{com}} - R_{\mathrm{lab}}| \leq 3.05 \times 10^{-5}$. The co-moving solver's solved-unknown speed reads $U = 1.08498$ against the laboratory solver's differenced 1.083. The modified volume identity of §3.3 holds to 7.2×10^{-8} relative

— which is the laboratory case's own volume drift on the same mesh, so the identity is satisfied to the accuracy at which conservation itself holds, and the window bookkeeping adds nothing measurable. The long-window demonstration completed: with $L = 40$ the window carried the tip to $R = 38.77$ at $t = 30$ ($u^* = 0.940$ and rising toward $\sqrt{2}$), more than three times the radius at which the laboratory annulus used in §5 exhausts its domain, at a per-step cost that does not grow with R — the structural property the window formulation exists to provide. The modified volume identity held to 3.5×10^{-7} relative over the whole run, and a second run at 1.6× coarser resolution reproduced $R(30)$ to 0.13%. The L-convergence ladder ($L = 20/40/80$) that would establish window-length independence is deliberately left to the follow-up campaign; $L = 40$ is asserted from the planar experience, not measured.

5 Validation

Three external comparisons, one end-to-end run each.

5.1 Reduction to the planar problem

At $R_0 = 100$ and $\mathrm{Oh} = 0.5$ the direct axisymmetric solver returns $u^*(t^* = 30) = 0.9309$ against the validated planar value 0.9297, a difference of 0.13% (figure 1(a)). The expected residual curvature effect at this radius is $O(h_0/R) \approx 1\%$ at the start and smaller as the hole grows, so a tenth of a percent is the right order and the right sign. This is the direct-solver counterpart of the depth-averaged gate in §4.3, and it exercises a completely different code path: a free surface with both curvatures, a moving mesh, remeshing, and the Newton tolerance law of §6.1.

5.2 Against a volume-of-fluid campaign

At coded $\mathrm{Oh} = 0.7071$ and $R_0 = 1$ the initial condition matches that of the 2024 volume-of-fluid campaign exactly, so no parameter matching is required. Over the record the two agree to 1.6% in the hole radius and 1.8% in the tip speed. The discrepancy is systematic rather than scattered, and it is quoted at face value: the reference's own resolution study is not recorded, so it cannot be attributed to either side. Appendix A reports a source-level audit whose purpose was to determine whether the difference could be a missing term on the finite-element side. It is not: every axisymmetric term the volume-of-fluid code adds by hand corresponds one-to-one with a term the finite-element tensor layer generates automatically, and the metric factors are exact. The ranked candidates that remain are all on the volume-of-fluid side. That reasoning is exhausted by an audit; it needs a measurement. The reference code was therefore rerun from scratch as a Newtonian axisymmetric case, carrying the three defect fixes recorded in §6 and an in-code tip diagnostic that the 2024 axisymmetric case never had, and compared against a finite-element leg at matched parameters. The rerun is not at the operating point of the comparison above: it is at $\mu = 0.05$, that is $\mathrm{Oh} = 0.0354$ through $\mu = \sqrt{2}\,\mathrm{Oh}$, with $R_0 = h_0 = 1$, carried to $t = 40$ ($t^* = 1600$, $R \simeq 52$) at `MAXlevel` 12.

Over that record the two methods agree to 0.51% in the hole radius, the largest departure anywhere after $t = 2$ (figure 2). The difference is not scatter: it grows smoothly from about 0.1% at $t = 4$, saturates by $t \approx 32$ and sits at -0.512% at the end of the record, with the volume-of-fluid radius consistently the smaller. A monotone, saturating offset of that size is what a diffuse interface and a real gas phase should produce against a sharp interface with no gas, and it is bounded by the interface thickness rather than growing with the solution.

Half a percent between two methods that share no discretisation, no interface representation and no time integrator is a much tighter agreement than the 1.6% and 1.8% above. Since the same volume-of-fluid method, freshly configured and free of the recorded defects, tracks the sharp-interface solver to that tolerance, the larger offset is unlikely to be a property of the method. The evidence therefore points at the 2024 campaign rather than at either code. It does not close

the question: the rerun is an order of magnitude lower in Ohnesorge number than the case it is being used to argue about, and a like-for-like rerun at Oh = 0.7071 is what would settle it. That run is cheap, and it is the obvious next measurement.

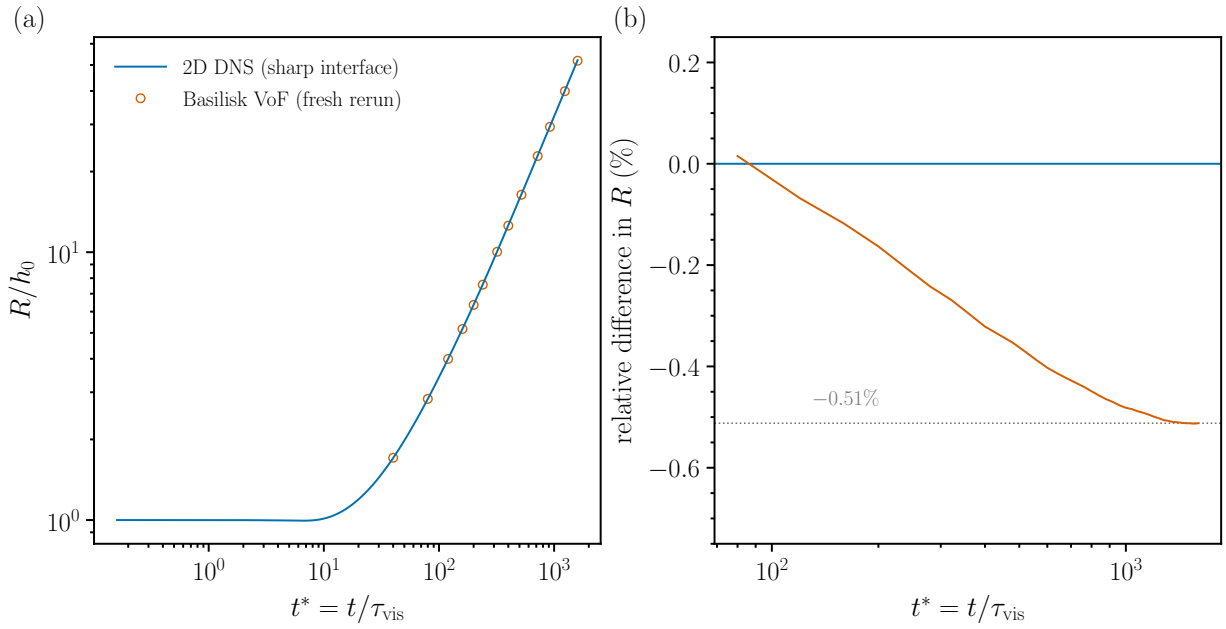


Figure 2: The fresh Newtonian axisymmetric rerun of the volume-of-fluid code against a finite-element leg at matched parameters, Oh = 0.0354 ($\mu = 0.05$), $R_0 = h_0 = 1$. (a) The hole radius over the whole retraction, from the cap-dominated start to the linear late-time growth. The volume-of-fluid record is written on a uniform $t = 1$ output grid, so its first point is at $t^* = 40$; the early transient is computed on both sides but sampled only by the finite-element leg. (b) The difference between them as a percentage of the finite-element radius, from $t = 2$. It is monotone and saturating rather than scattered, reaching −0.512% at the end of the record, with the volume-of-fluid radius the smaller throughout — the signature of a diffuse interface and a resolved gas phase against a sharp interface with none, not of a disagreement that grows with the solution.

At $R_0 = 1$ the direct solver shows a mesh-independent feature that the depth-averaged model cannot represent at all. The tip does not open monotonically: the hole radius *closes* by 0.49%, reaching its minimum near $t = 0.15$, before turning around and growing smoothly with $\dot{R}$ climbing towards V_{TC}. The shape is identical at two resolutions differing by a factor of four in near-tip element count, and it is the interface–symmetry corner node itself that recedes. The mechanism is the one the tip correction (8) describes: at $R \sim h_{\mathrm{tip}} \sim 1$ the toroidal term is $O(1)$ against the leading −2, the net driving force is genuinely weak, and at low Ohnesorge number an impulsively started flow can transiently go the other way. The proximity to the Taylor & Michael closing threshold [7] is not incidental. A quasi-static massless-cap closure, with either constant, has no way to encode "weak enough to be beaten by inertia for a while" and so cannot produce this at all.

5.3 The depth-averaged model against the direct solution

The comparison at $R_0 = 1$ requires care with the initial condition. The direct solver's cap of radius 1/2 carries volume that the clean-hole depth-averaged initial condition does not, so the reduced model is started at the volume-matched radius

$$R_{\mathrm{eq}}^2 = r_s^2 - \frac{\pi a r_s}{2} + \frac{2a^2}{3}, \qquad r_s = R_0 + a, \quad a = \tfrac{1}{2}, \tag{9}$$

giving $R_{\mathrm{eq}} = 1.1129$ for $R_0 = 1$. The quantity compared is $\dot{R}$ and not R: the depth-averaged tip is the matched edge of an asymptotic construction while the direct hole radius is the innermost interface point, and the offset between them is an irreducible $O(h/2)$.

At $\mathrm{Oh} = 0.7071$ and $t^* = 100$,

	$u^*(t^* = 100)$
direct solver (AXI-LAB)	0.9613
TC1D-AXI-ALE, default closure $N_{rr}(R) = -2$	0.9966
TC1D-AXI-ALE, with toroidal correction (8)	0.8976

The two closures bracket the direct solution, from +3.7% above and −6.6% below (figure 3). Neither is right, and that is the result: at $R_0 \sim 1$ the azimuthal curvature contributes an $O(1)$ fraction of the edge force, so the leading-order closure omits a term of its own order, while the first correction (8) overshoots it. The discrepancy is located in the edge condition and not in the bulk operators, which is exactly what bracketing establishes and what a single closure could not. The sign of the default closure's error is also informative: its overshoot is the same planar-style bias the reduced model shows against the direct solution in the planar geometry, so it is inherited rather than new.

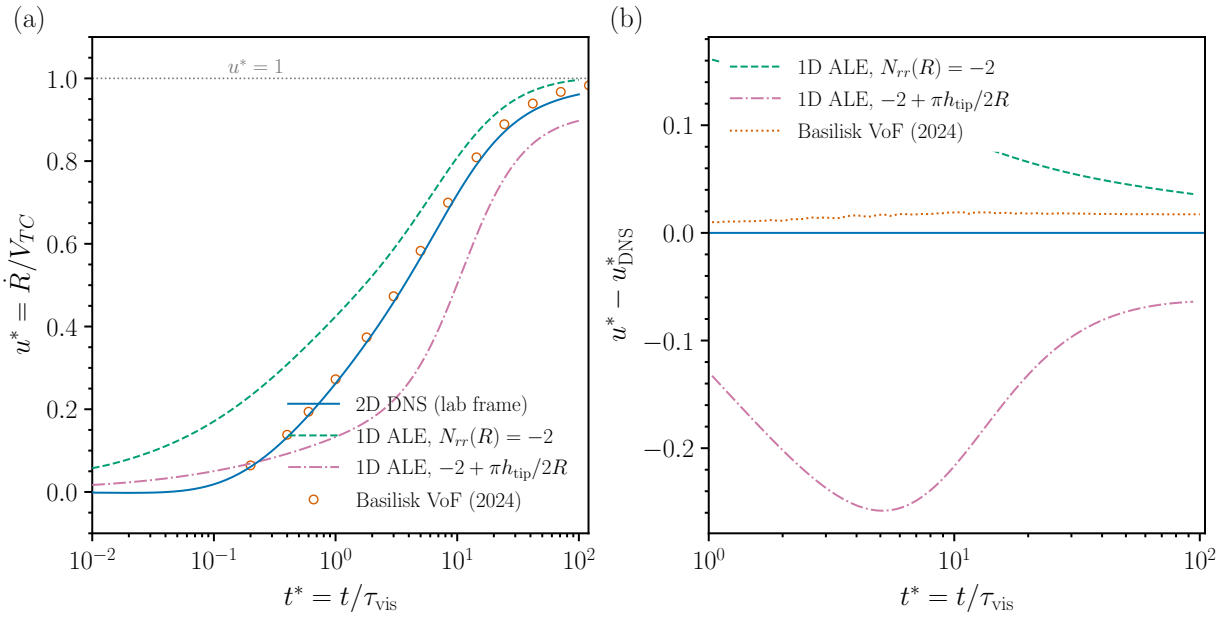


Figure 3: The two tip closures bracket the direct solution at $\mathrm{Oh} = 0.7071$, $R_0 = 1$. (a) Retraction speed: solid, direct solver in the laboratory frame; dashed, TC1D-AXI-ALE with the default closure $N_{rr}(R) = -2$; dash-dot, TC1D-AXI-ALE with the toroidal correction (8); circles, the 2024 volume-of-fluid campaign. The depth-averaged runs start at the volume-matched radius (9). The direct record is genuinely negative over the first ~ 0.2 viscous times — the tip recedes before it opens (§5.2) — and is drawn on linear ordinates so that it is visible rather than eaten by a logarithmic axis. (b) Departure from the direct solution, on the direct solver's own abscissa; the other three records are interpolated onto it, which is safe here because all four are smooth and monotone over the window shown. The two depth-averaged curves lie on opposite sides of zero over the whole record.

The mass-coordinate solver was run in the same configuration as a cross-check. Its first attempt terminated in the Newton floor of §6.1 — the observation that motivated porting the $\varepsilon\, r_{\max}^2/\Delta t$ tolerance rule to that solver — and the repaired rerun advanced well past the original failure times ($t^* = 73$ and 88 of 100) but was stopped short of completion as outside this strand's

scope. It is therefore omitted from figure 3 rather than plotted short. The two solvers agree to 7.3×10^{-6} over the range where both are available (§4.2), so nothing in the conclusion turns on it.

Two of Savva & Bush's axisymmetric curves were digitised with the same calibrated workflow as the planar figures (`validation/data/`, with per-figure calibration overlays): their figure 16 inset curve C ($\mathrm{Oh_{SB}} = 1$, $R_0 = 50H$ — at this Ohnesorge number their viscous and inertial clocks coincide, so the comparison is convention-proof) and figure 10 curve G ($\mathrm{Oh_{SB}} = 10$, $R_0 = 50H$, viscous clock). Against the mass-coordinate solver at matched R_0: at $\mathrm{Oh_{SB}} = 1$ over $t^* \in [2, 10]$ the default closure agrees to 1.9% rms ($u^*(10) = 0.9133$ against their 0.8985), and the toroidal tip correction *improves* this to 1.3% rms — their model resolves the smoothed rim cap with full capillarity, which is exactly the physics the correction reinstates at leading order. At $\mathrm{Oh_{SB}} = 10$ over $t^* \in [2, 34.5]$ the agreement is 3.5% rms with the deviation concentrated in the early transient (where their published sensitivity to the initial tip-smoothing parameter α is strongest, their figure 14) and narrowing to 1.8% by $t^* = 34.5$. Digitisation accuracy is 0.15–0.3% of full scale. A third digitised curve (their figure 16 main panel, curve D) is retained in the data directory but excluded here: its time-axis normalisation in the moderate-Oh section could not be established beyond doubt, and a comparison built on a guessed clock would be worse than none. The self-similar collapse at matched $\mathrm{Oh}\, h_0/R_0$ remains the natural test to build on these curves.

6 Four numerical traps

Each of these cost real time, and three of them present as something other than what they are. They are recorded with the evidence that discriminated the cause, because in every case the plausible first diagnosis was wrong.

6.1 The Newton absolute-tolerance floor

Symptom. A run at large radius aborts inside its first output interval with residuals oscillating at $O(0.1)$ and the timestep controller halving until it refuses to go further. The oscillation looks structural and looks independent of the timestep, and is neither.

Cause. The solver's convergence test is an absolute bound on the max-norm of the residual vector, and the axisymmetric coordinate system multiplies every residual entry by $2\pi r$. The residual magnitude of the same physical imbalance therefore grows linearly with the hole radius. A direct solve cannot drive a residual below roughly $\varepsilon \|J\| \, \|u\|$; here $\|u\|$ is dominated by mesh-position unknowns, which are absolute radii $\sim R$, and $\|J\|$ by the mass rows, which carry the measure weight $\sim R$ divided by the timestep. Hence a floor $\sim \varepsilon R^2/\Delta t$. Every rejected step halves Δt, which *raises* the floor, so the cascade is self-reinforcing and the $O(0.1)$ oscillation is its tail rather than its cause.

Discriminating evidence. A one-variable-at-a-time scan in R_0 at identical mesh, unknown count and timestep gives initial residuals exactly linear in R_0 and final residuals 2.8×10^{-11}, 5.3×10^{-10}, 7.6×10^{-9} and 3.7–5.2×10^{-8} at $R_0 = 1$, 10, 30 and 100; only the last never stops, so only the last measures a floor rather than bounding it. A smooth onset that tracks the coordinate offset is a scaling, not a defect that switches on. Independently, a timestep scan at fixed $R_0 = 100$ gives floors from 9.2×10^{-10} at $\Delta t = 6.8 \times 10^{-4}$ to 2.0×10^{-6} at $\Delta t = 3.4 \times 10^{-7}$, tracking $\varepsilon R^2/\Delta t$ to within a factor of about five throughout. What it is *not* was checked in each case: not a residual or Jacobian error (the finite-difference Jacobian is clean at $R_0 = 100$, and the identical configuration converges to 6.1×10^{-10} at a larger timestep), not the graded mesh (the failure reproduces on a uniform one), not the Newton budget (the residual sits at the same value for a further thirty-one iterations, and a line search does not move it), and not the time-integrator start-up.

Remedy. Retune the tolerance per timestep as $\max(\varepsilon_0, \, s\,\varepsilon R^2/\Delta t)$ with a safety factor s, which holds roughly a thirtyfold margin over the measured floor uniformly in Δt. A fixed tolerance is

a lottery, because the last sub-step of an output interval is a leftover sliver whose floor may be orders of magnitude above the interval's. Rescaling the test space by $1/R$ instead would move one of the two factors into the residual and leave the Newton step identical; it would not touch the $1/\Delta t$ factor, which is the one that kills runs. In the window formulation the scale is read live from the global position unknown rather than from the initial radius, since R is no longer approximately constant.

This trap appeared three times in this strand, with three different scales: R_0^2 in the laboratory solver, $(X + L)^2$ in the window solver, and $R_0^2 + 2M$ in the mass-coordinate solver, where the domain is a mass and the largest radius is $\sqrt{R_0^2 + 2M}$. The last of these is why the mass-coordinate legs of §5.3 are short: at the comparison's domain size the floor is three orders of magnitude above what the check suite's shorter horizons ever see. The depth-averaged ALE solver meets the same mechanism in a milder form — Newton lands on a genuine fixed point, or a two-cycle between two bit-identical values, at a few times 10^{-8} and errors out only because that is just above tolerance — and is cured by a fixed looser absolute tolerance, still nine orders tighter than the velocity scale of the physics, verified over $R_0 \in \{5, 200, 1000\}$ and three decades of starting step.

The underlying absolute test is inherited from the upstream library [9]; no relative criterion exists there either, so the per-step retune is the correct local remedy and a relative-tolerance option is upstream work.

6.2 A double Lagrange multiplier at the outflow corner

Symptom. In the window formulation the linear solver reports a backward error of order 10^{47}, the first Newton step returns a non-finite value, and every timestep is rejected until the controller refuses to reduce further. That last part reads exactly like the cascade of §6.1 and is a completely different failure. A finite-difference Jacobian check does not catch it: the assembled Jacobian is correct, and it is the system that is rank-deficient.

Cause. At the corner where the outflow boundary meets the free surface, the axial mesh coordinate is pinned but the radial one is a free unknown, and two Lagrange multipliers act on that one unknown — the free surface's kinematic multiplier and the multiplier enforcing the window's far edge. Two constraints on one unknown is a null space. The laboratory case never sees it, because an ordinary Dirichlet condition removes the position unknowns and the framework then disposes of the orphaned kinematic multiplier itself.

Discriminating evidence. The singular values of the assembled Jacobian, smallest first, are 4.8×10^{-13}, 2.3×10^{-12}, 4.6×10^{-4} as written; pinning the corner multiplier removes the first and leaves 2.3×10^{-12}, 4.6×10^{-4}; the laboratory case on the same mesh gives 2.3×10^{-12}, 4.7×10^{-4}. The largest singular value is 5.3×10^{3}, and there are no zero rows or columns, so the deficiency is structural rather than a missing entry. The offending mode is entirely in the kinematic multiplier, dominated by a single node with an exponentially decaying tail along the interface — the signature of one redundant constraint, not of a physical soft mode. The mode that survives is not introduced here at all: the laboratory case carries the identical mode at the same magnitude. Pinning the corner multiplier therefore restores the working case's conditioning exactly, and nothing more.

Two further points of the same formulation are worth recording because each was initially written the natural way and was wrong. The tip-pinning constraint must be written in residual form as $\texttt{mesh_x} - X$ rather than as $\texttt{mesh_x}$: the tip sits at $r \sim R_0 > 0$, so under the $2\pi r$ point measure the constraint $2\pi r_{\mathrm{tip}}\, \texttt{mesh_x}_{\mathrm{tip}} = 0$ is a double root whose derivative vanishes at the solution, giving a singular Jacobian row, whereas $\texttt{mesh_x} - X$ has derivative -1 identically. And that pin must be evaluated in a Cartesian measure: it is a geometric statement that a node sits at a radius, not an integral of anything, and the $2\pi R(t)$ factor an axisymmetric measure would attach to it grows without bound over a long window run.

6.3 An argument-parser prefix collision

The framework parses the real command line at problem initialisation, with prefix matching active, and one of its own flags is `--outdir`. A script's own `--out` option is therefore accepted as that flag and silently overrides every per-case output directory in the process. There is no error and no warning; runs simply write somewhere else. The remedy in the drivers is to suppress command-line parsing outright; in the check scripts, which also invoke the planar reference through a driver that does not suppress it, the argument vector is emptied before the first solver is constructed. This is recorded because the symptom — output appearing in one directory when several were requested — reads as a bug in the script's own path handling.

6.4 An unbounded domain estimate

The small-hole exponential law $R = R_0 \exp(t^*/2)$ is an *early-time* statement: it holds only while the edge speed $R/(2\tau_{\mathrm{vis}})$ is below V_{TC}, that is until R reaches the crossover radius $\sqrt{2}\mu$, past which the edge cannot outrun the Taylor–Culick speed. Sizing a domain from the raw exponential at a large end time therefore requests an astronomically long mesh — of order 10^{21} at $t^* = 100$ — and the resulting mesh coordinates blow the first Newton solve up to residuals of order 10^{34}, which is how the bound was found. The driver now computes both the exponential estimate, capped at the crossover radius, and the linear estimate $V_{TC}t$, and takes the larger: no single formula covers the whole range of R_0^*. The estimate is a starting point and not a promise, and every output leg checks the far-field drift.

A related sizing point is worth stating because the planar rule looks transferable and is not. In the mass coordinate the domain is $M = \int_R^{r_{\max}} rh\,\mathrm{d}r$, which for an undisturbed film is $(r_{\max}^2 - R_0^2)/2$ — quadratic in the radial extent. Reusing the planar expression, which returns a length, sizes the domain by a factor of order $r_{\max}$ too small.

7 Limitations, and what would remove them

Window-length convergence is not established. The window formulation has been verified against the laboratory solver over an early window (§4.4), which tests that the two agree where both are valid. It has not been run long enough for the semi-infinite behaviour it exists to capture, and the window length L has not been varied. Until an L-ladder exists, a long window run reports a length as much as a physics.

The smoke ladder does not cover large R_0. The quick physics probe runs at $R_0 = 1$; the planar-reduction case at $R_0 = 100$ is a multi-hour comparison and is deliberately not a smoke test. The consequence is that the Newton tolerance law of §6.1 is exercised by the campaign rather than by the fast checks, and its scale is set from the *initial* radius rather than the live one. Over the runs reported here R grows by about 15%, so the distinction does not bite; a campaign in which R grows by an order of magnitude from a large R_0 should re-measure rather than assume.

Remeshing costs volume conservation four orders of magnitude. Drift is 2.1×10^{-9} over 200 steps without remeshing and about 2.6×10^{-5} over a full run with it. The difference is interpolation at the remesh, not accumulation between them, so it is bounded by the number of remeshes rather than by the run length — but it does set the floor on what the volume identity can certify, and it is the reason the window's modified identity is quoted as matching the laboratory case's own drift rather than as being exact.

The tip closure is bracketed, not determined. §5.3 shows that the leading-order closure and its first geometric correction lie either side of the direct solution at $R_0 \sim 1$. That localises the error and does not fix it. Two routes are open: a systematic $O(h/R)$ matched-asymptotic treatment of the toroidal rim, which would replace (8) with a derived rather than an estimated coefficient; and an inertial cap, which is what the transient reversal of §5.2 indicates is missing, since a

quasi-static massless cap cannot produce it at any coefficient. The second is the more interesting because it is a statement about the model rather than about its constant.

The comparison set is thin by design. One run per question, no sweep in Oh or R_0, and no digitised axisymmetric literature curves. The verification evidence of §4 is meant to carry the weight that a sweep would otherwise carry, and it is worth being explicit that it cannot carry all of it: internal consistency at high precision is not external agreement.

A Two source-level audits

Both were run after the validation gates of §5 had passed, to answer a question the gates raise but cannot settle: whether the residual disagreement with the volume-of-fluid reference could be a term missing on the finite-element side.

Against the volume-of-fluid code's axisymmetric module. Every axisymmetric term that code adds by hand — the hoop-viscosity patch in its viscous solver, the three separate azimuthal-curvature additions in its curvature routine, and, for the viscoelastic extension, a bespoke hoop-stress field with its own momentum forcing — corresponds one-to-one with a term the finite-element tensor layer produces automatically: the u_r/r entry of the velocity gradient, the $(T_{rr} - T_{\theta\theta})/r$ entry of the tensor divergence, and the surface divergence. No physics is missing on the finite-element side, and the metric factors themselves are exact and are ruled out as the source of the 1.6–1.8% offset. The remaining candidates, ranked and all on the volume-of-fluid side, are height-function curvature noise together with the explicit floor that method places on the azimuthal curvature; the hoop viscous term being an acknowledged after-the-fact patch; adaptive coarse/fine transitions at the rim; and the resolved ambient phase that a single-phase formulation excludes. A useful corollary for the viscoelastic extension: the polymeric hoop stress will arrive through the tensor divergence with no bespoke module.

Against the underlying library's hand-written axisymmetric elements. The residuals generated for AXI-LAB are term-for-term identical to the library's axisymmetric Navier–Stokes element — momentum including the u_r/r^2 hoop viscous and $p\psi/r$ hoop pressure terms, the continuity radius factors, the ALE corrections and the stress-divergence form — and to its axisymmetric interface element, including the azimuthal-curvature injection in the surface-divergence capillary term, up to a global factor of 2π and a global sign [9]. The one omitted end-of-interface co-normal term lands entirely in pinned equations at both ends, so it is discarded correctly rather than lost. This audit is also where the absolute Newton tolerance of §6.1 was traced upstream.

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